

Eco-Evolutionary Rescue and Spatial Bistability in a Roy-Style Predator–Prey Model

Mathbio research notes

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Abstract

The fixed-defense spatial Roy-style model did not produce robust predator rescue: the best fixed-defense spatial candidate failed validation and was classified as a transient or numerical candidate. Introducing prey defense evolution in the well-mixed model did support indirect evolutionary rescue, increasing the predator mortality stress threshold from the no-evolution value to the evolving threshold while defense frequency declined in the rescue window. Adding spatial diffusion did not produce a result that can be summarized by one scalar persistence threshold. Later diagnostics showed classifier sensitivity, unresolved long transients, continuation hysteresis, and bistability. The current endpoint is that ODE prey defense evolution supports indirect evolutionary rescue, while the spatial PDE response is path-dependent and bistable. Basin entry depends on initial defense frequency q_0 and initial predator abundance w_0 , so the spatial model should be analyzed through basin structure rather than a single persistence threshold.

1 Introduction

Indirect evolutionary rescue requires a change in trait values or genotype frequencies after environmental stress. A fixed-defense predator–prey model can test whether spatial patterning alone supports predator persistence, but it cannot test indirect evolutionary rescue because the prey defense state is fixed. The eco-evolutionary extension therefore introduces a prey defense frequency q , where lower q represents lower defense, higher prey growth, and higher predator conversion opportunity.

Spatial structure can change predator–prey outcomes by changing local overlap, covariance, and transient recovery paths. The current project initially asked whether spatial diffusion amplifies or suppresses the well-mixed evolutionary rescue threshold. That framing proved too narrow. The first threshold comparison suggested a lower PDE-evo threshold than the ODE-evo threshold, but later analyses showed that threshold language is insufficient because the system has multiple reachable outcomes at the same stress.

The report summarizes the current endpoint from existing project outputs. It does not introduce new simulations or model equations. The central conclusion is that the spatial eco-evolutionary model has path-dependent bistable rescue dynamics organized by reachable outcome basins.

2 Model sequence

The eco-evolutionary variables are total prey density n , predator density w , prey defense frequency q , and free space

$$z = \kappa^{-1} - n - w.$$

The defense trade-off functions are

$$r(q) = r_u(1 - q) + r_v q, \quad a(q) = a_u(1 - q) + a_v q, \quad b(q) = b_u(1 - q) + b_v q.$$

The well-mixed ODE is

$$\frac{dn}{dt} = n[r(q)z - \xi - a(q)w], \quad (1)$$

$$\frac{dw}{dt} = w[b(q)nz - (m + s) - \mu w], \quad (2)$$

$$\frac{dq}{dt} = \nu q(1 - q)[(r_v - r_u)z - (a_v - a_u)w]. \quad (3)$$

Here s is the predator mortality stress. The spatial PDE uses the same local reaction terms and adds diffusion:

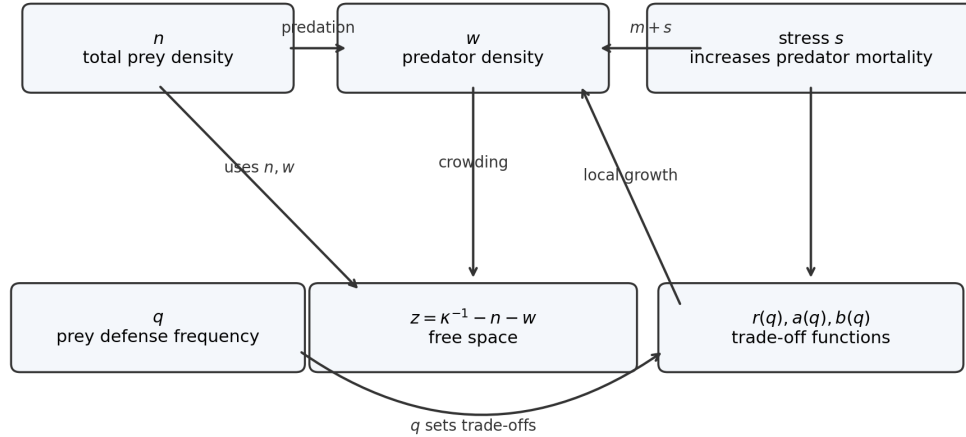
$$\partial_t n = D_n \nabla^2 n + n[r(q)z - \xi - a(q)w], \quad (4)$$

$$\partial_t w = D_w \nabla^2 w + w[b(q)nz - (m + s) - \mu w], \quad (5)$$

$$\partial_t q = D_q \nabla^2 q + \nu q(1 - q)[(r_v - r_u)z - (a_v - a_u)w]. \quad (6)$$

The PDE simulations used zero-flux Neumann boundaries.

Roy eco-evolutionary rescue model variables



ODE uses spatial means; PDE adds diffusion of n , w , and q with zero-flux boundaries.

Figure 1: Conceptual schematic of the eco-evolutionary variables and trade-off links.

3 Fixed-defense spatial model

The fixed-defense spatial Roy-style route tested whether spatial patterning alone could rescue predators when the well-mixed reference failed. That route did not produce robust ODE-extinct/PDE-persistent rescue. The best weak candidate occurred at $D_w/D_u = 150$, but it failed the validation sequence and was classified as **transient_or_numerical_candidate**. Therefore fixed-defense spatial patterning was not enough for a robust predator-rescue claim in the tested model.

Table 1: Parameter values used in the eco-evolutionary tests. The conversion values $b_u = 0.08$ and $b_v = 0.02$ were used to produce an interior defense-frequency baseline.

Symbol	Value	Description
κ	0.15	free-space scale
ξ	0.55	prey loss term
r_u	1	undefended prey growth
r_v	0.65	defended prey growth
a_u	1	undefended predation coefficient
a_v	0.35	defended predation coefficient
b_u	0.08	undefended conversion coefficient
b_v	0.02	defended conversion coefficient
m	0.1	baseline predator mortality
μ	0.2	predator density dependence
ν	0.05	evolutionary rate
D_n	0.01	prey diffusion
D_w	0.01	predator diffusion
D_q	0.005	defense-frequency diffusion

4 Eco-evolutionary ODE rescue

The well-mixed eco-evolutionary ODE supported indirect evolutionary rescue. The no-evolution predator mortality stress threshold was

$$m_c^{ODE, \text{no evo}} = 0.069448242,$$

while the evolving threshold was

$$m_c^{ODE, \text{evo}} = 0.16486816.$$

The evolutionary rescue effect was therefore

$$\Delta_{evo}^{ODE} = 0.095419922.$$

In the ODE rescue window, prey defense frequency declined from approximately 0.6726 to approximately 0.3336, consistent with reduced defense and increased palatability supporting predator recovery.

Table 2: Threshold quantities from the ODE rescue analysis and early PDE screening. The PDE entries are historical screening quantities and should not be interpreted as final scalar thresholds.

Quantity	Value	Interpretation
$m_c^{ODE, \text{no evo}}$	0.0694482422	ODE reference
$m_c^{ODE, \text{evo}}$	0.164868164	ODE reference
Δ_{evo}^{ODE}	0.0954199219	ODE evolutionary rescue effect
representative $m_c^{PDE, \text{no evo}}$	0.06921875	historical screening quantity
representative $m_c^{PDE, \text{evo}}$	0.11765625	historical screening quantity

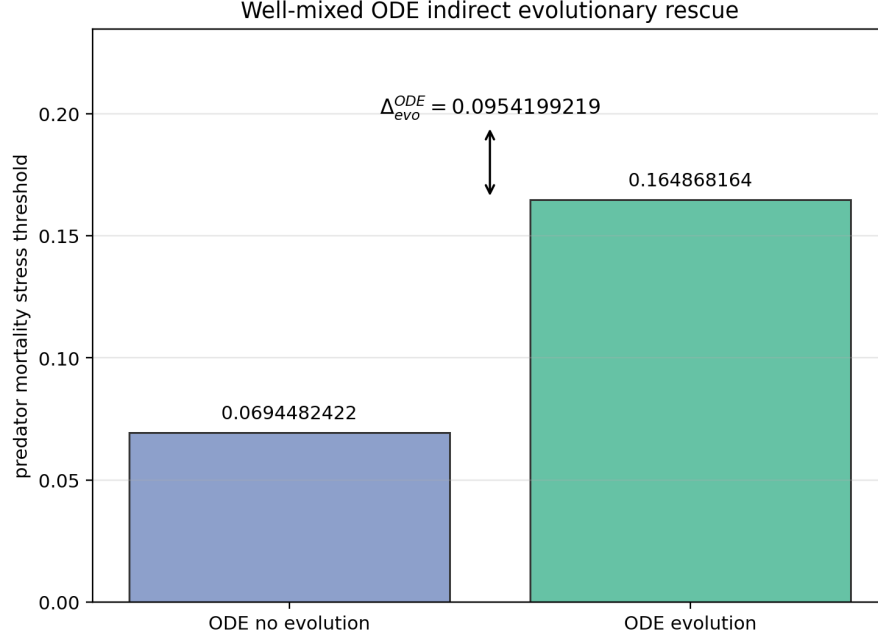


Figure 2: ODE threshold comparison showing indirect evolutionary rescue in the well-mixed model.

5 Spatial PDE and failure of scalar threshold language

The first spatial threshold comparison reported a representative PDE no-evolution threshold of 0.06921875 and a representative PDE evolution threshold of 0.11765625. Because the latter is below the ODE evolution threshold 0.16486816, the initial screening suggested a lower PDE-evo persistence threshold than the ODE-evo threshold.

That interpretation was later corrected. The diagnostic sequence found:

- PR #4: `pde_evo_threshold_classifier_sensitive`;
- PR #5: `pde_evo_persistence_unresolved`;
- PR #6: `pde_evo_hysteresis_detected`;
- PR #7: `pde_evo_bistability_mapped`;
- PR #9: `basin_boundary_mapped`.

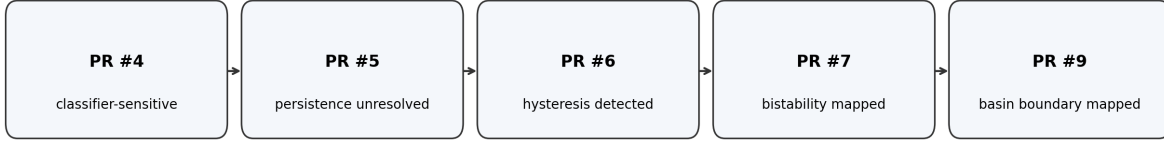
Together these results show that the spatial PDE does not behave like a single monotone stress-response curve in the sensitive stress range.

6 Hysteresis and basin structure

Continuation analysis showed direction dependence: upward continuation could keep a predator-persistent branch at stresses where downward continuation could remain extinct or transient. Basin mapping then showed that direction dependence was not merely a numerical artifact of a continuation path. Different initial-condition families could reach different outcomes at the same stress.

The Step 13 basin map found persistent/transient mixed regimes at 0.141262205 and 0.15, and bistable persistent/extinct regimes at 0.1584375, 0.16486816, and 0.175.

Diagnostic sequence correcting the initial threshold narrative



The spatial PDE response is path-dependent, so scalar threshold language is insufficient.

Figure 3: Diagnostic sequence that replaced the early scalar-threshold interpretation with a path-dependent basin interpretation.

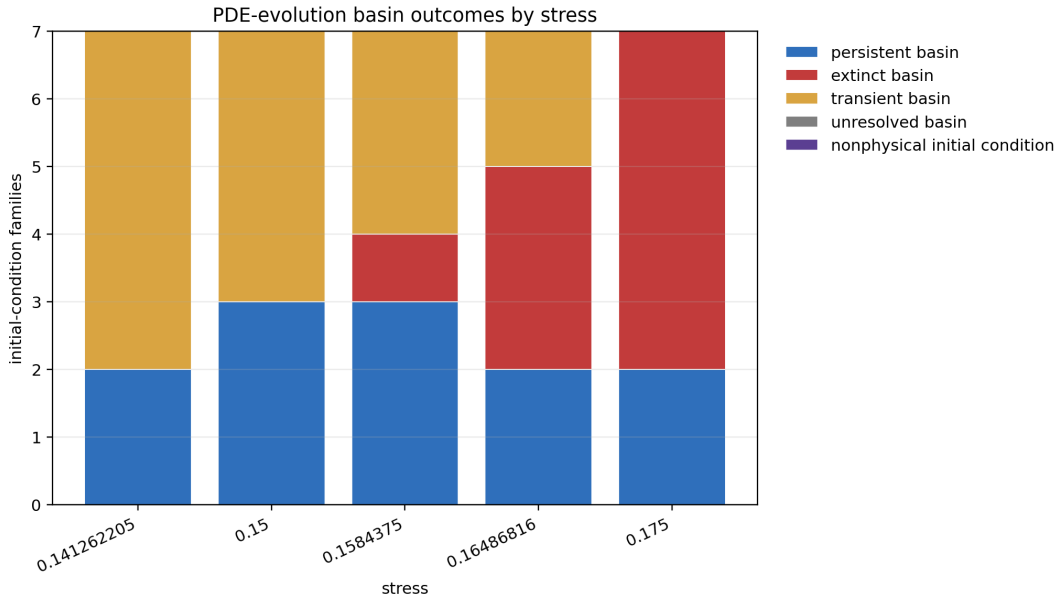


Figure 4: Basin outcomes across the Step 13 initial-condition families. Bistability appears where persistent and extinct basin outcomes coexist.

Table 3: Stress-regime map from the bistability analysis.

Stress	Regime
0.141262205	<code>persistent_transient_mixed</code>
0.15	<code>persistent_transient_mixed</code>
0.1584375	<code>bistable_persistent_extinct</code>
0.16486816	<code>bistable_persistent_extinct</code>
0.175	<code>bistable_persistent_extinct</code>

7 Basin-boundary dependence on q_0 and w_0

PR #9 quantified basin boundaries by scanning initial defense frequency q_0 and predator abundance scale w_0/w_{baseline} at two stresses inside the bistable interval. Both target stresses were classified as `bistable_persistent_extinct`. At stress 0.1584375, the grid contained 20 persistent, 7 extinct, and 43 transient outcomes. At stress 0.16486816, it contained 14 persistent, 21 extinct, and 35 transient outcomes.

Persistent basins concentrated at low-to-intermediate q_0 . Predator initial abundance modulated basin entry within that range: some initial predator scales entered persistence, while neighboring points remained transient or entered extinction. Transient outcomes remain common, so the current basin boundary is interpretable but not fully sharp at the present horizon.

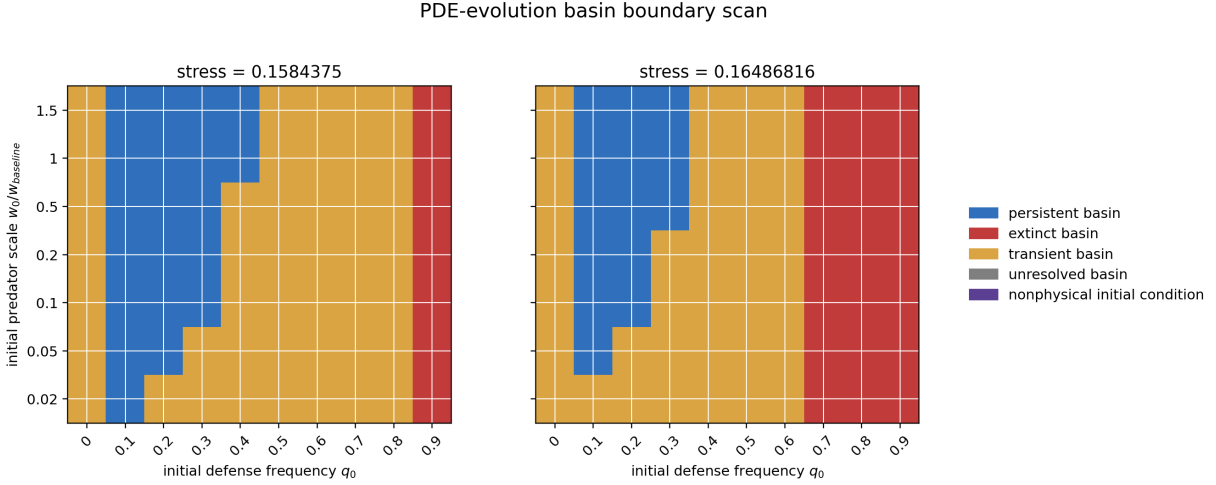


Figure 5: Basin-boundary scan over initial defense frequency q_0 and predator abundance scale w_0/w_{baseline} .

Table 4: Basin-boundary counts from the q_0 – w_0 scan.

Stress	Persistent	Extinct	Transient	Unresolved	Nonphysical	Regime
0.1584375	20	7	43	0	0	<code>bistable_persistent_extinct</code>
0.16486816	14	21	35	0	0	<code>bistable_persistent_extinct</code>

8 Discussion

The fixed-defense route did not support robust spatial rescue, while the well-mixed eco-evolutionary route did support indirect evolutionary rescue. The spatial eco-evolutionary route initially looked like a threshold-suppression result, but the subsequent diagnostic sequence changed the interpretation.

The current result is not a simple spatial suppression result. It is a bistability and basin-structure result. The PDE can reach predator-persistent or predator-extinct outcomes at the same stress, depending on initial condition or continuation path. The most recent basin-boundary scan shows that q_0 and w_0 are important basin-entry coordinates.

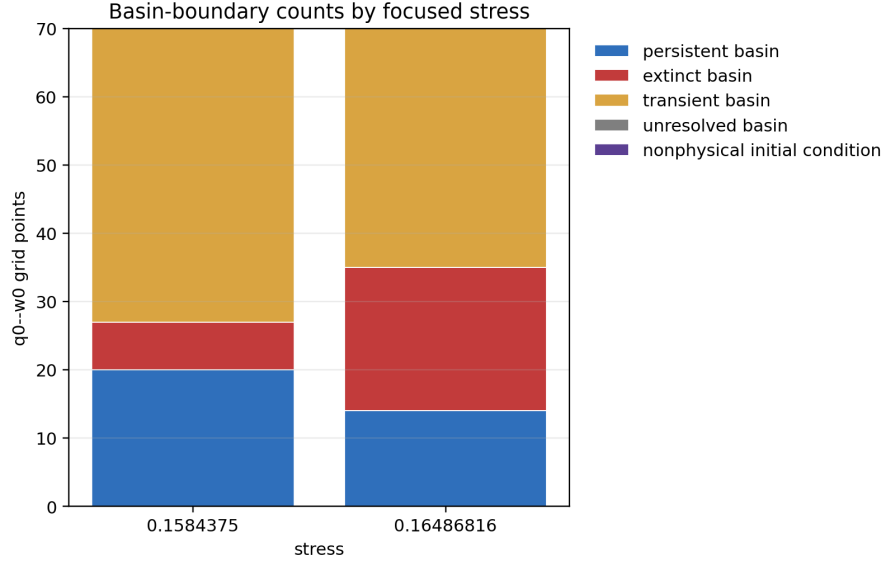


Figure 6: Outcome counts in the focused PR #9 basin-boundary scan.

The current analysis has several limitations. Transient outcomes remain common in the q_0-w_0 grid, so the separatrix is not yet fully resolved. Diffusion coefficient sensitivity remains untested after basin mapping. The report also summarizes one trade-off parameterization, with $b_u = 0.08$ and $b_v = 0.02$, rather than a broad family of trade-offs.

9 Current conclusion and next work

ODE prey defense evolution supports indirect evolutionary rescue. In the spatial PDE, the evolutionary rescue response is path-dependent and bistable. Basin entry depends on initial defense frequency and predator abundance, so the spatial model should be analyzed through basin structure rather than a single persistence threshold.

The next simulation step is adaptive refinement of the q_0-w_0 basin boundary for stresses 0.1584375 and 0.16486816.